Eigenvalues and inverse problems

The sharp generic threshold for spectral-subspace rotation

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging reflects a sharp perturbation threshold without geometric separation assumptions beyond a spectral gap; community impact is the stability of spectral subspaces.

Let A be a self-adjoint, possibly unbounded operator on a separable complex Hilbert space. Suppose $\sigma(A) = \sigma \cup \Sigma$, where the two nonempty closed sets satisfy $d = \operatorname{dist}(\sigma, \Sigma) > 0$. For a bounded self-adjoint perturbation V, set

$$P = E_A(\sigma), \qquad Q = E_{A+V}(O_{d/2}(\sigma)), \qquad O_r(S) = \{x \in \mathbb{R} : \operatorname{dist}(x, S) < r\},$$

where E_B denotes the spectral projection measure of B.

Does $\|V\| < d/2$ always imply $\|P - Q\| < 1$? All norms are operator norms. Equivalently, is the optimal universal constant $c_{\text{opt}} = 1/2$ for guaranteeing that the maximal angle $\arcsin \|P - Q\|$ is strictly below $\pi/2$?

The two spectral sets may interlace: no ordering or disjoint-convex-hull hypothesis is imposed. The source's operator formulation is retained; Hermitian matrix invariant-subspace perturbation is its finite-dimensional setting. This problem concerns subspace orientation after perturbation, beyond preservation of a spectral gap.

References

A. Seelmann, *Notes on the subspace perturbation problem for off-diagonal perturbations*, Proceedings AMS 144 (2016), 3825–3832, introduction p.3826, the paragraph on general perturbations (primary manuscript; journal). Seelmann, *On an estimate in the subspace perturbation problem*, Journal d'Analyse Mathématique 135 (2018), 313–343 (primary manuscript). Seelmann, *Unifying the treatment of indefinite and semidefinite perturbations in the subspace perturbation problem*, Operators and Matrices 15 (2021), 1181–1188, Theorems 1.1–1.2 (primary paper).

Status check — 2026-09-08

The 2021 generic theorem retains the sufficient constant $c_{\text{crit}} = 0.4548399\ldots$; its sharp theorem at a larger threshold assumes extra spectral separation. The 2018 paper's solved optimization problem determines an upper angle bound, not the conjectured threshold. Searches for "subspace perturbation", "optimal constant", "1/2", "Seelmann", "conjecture", and 2024–2026 found no resolution. No recent explicit reaffirmation of the exact endpoint conjecture was located; the status evidence is bounded.

Audit update — 2026-09-10

Rechecked Seelmann's primary account and the 2021 perturbation paper. The generic sharp threshold remains distinct from the established sufficient constants and stronger results under extra geometry. Searches for subsequent optimal generic subspace-rotation thresholds found no solution; these non-sharp estimates do not complete the displayed optimum.